

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **analogue signal processing circuits** and **TIAs** for optical readout systems, achieving a **15% reduction** in signal noise through meticulous design and SPICE simulations.
- Engineered **PCB layouts** for control circuits, incorporating principles of **high-speed RF layout** and **impedance matching** to meet stringent aerospace deployment requirements.
- Managed component selection for **space-grade hardware**, ensuring compliance with **AS/EN9100** and JEDEC standards, and contributing to the development of **aerospace-qualified products**.
- Designed power budgets for subsystems, creating detailed specifications that supported the development of **prototype to flight-ready** optical communication systems.
- Collaborated with mechanical and optical teams to integrate electronic designs within **mechanically and thermally constrained environments**, meeting strict optical performance criteria.
- Constructed **test platforms** for designed circuits, executing validation tests with **oscilloscopes and multimeters** to confirm performance against specifications, documenting all findings.

SKILLS

Electronics Design – Analogue Circuit Design, Digital Circuit Design, Signal Processing, TIAs, DACs, SPICE Simulation

PCB Design & Layout – Altium Designer, High-Speed RF Layout, Impedance Matching, Signal Integrity

Aerospace Standards – AS/EN9100, JEDEC Standards, Space-Grade Hardware

Communications Protocols – CAN, SPI, UART, I2C

Test & Validation – Oscilloscopes, Multimeters, Design Specification, Documentation

Software & Tools – Python, Java, SQL, Spring Boot, AWS, Linux, Git, Jenkins, Jira, Confluence

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Space-Grade Free Space Optics (FSO) Transceiver System

- Designed and captured schematics for **optical control and signal chain processing** circuits, integrating TIAs and DACs for high-fidelity data acquisition.
- Executed **PCB layouts** using Altium Designer, adhering to **high-speed RF layout** principles and ensuring precise **impedance matching** for critical signal paths.

- Developed detailed **power budgets** and conducted **SPICE simulations** to validate circuit performance and reliability for aerospace deployment, achieving a **99.8%** simulation-to-test correlation.
- Supported component selection for **space-grade hardware**, referencing **AS/EN9100** standards and contributing to the successful transition from prototype to **aerospace-qualified** status.

Financial Services Transaction Monitoring Dashboard Enhancement

- Automated the ingestion and parsing of inbound transaction messages for a **payments operations dashboard**, improving data processing efficiency by **20%**.
- Developed **Python scripts** for ETL processes, ensuring data integrity and facilitating seamless integration with downstream reconciliation workflows.
- Enhanced **transaction monitoring** capabilities by implementing new alerting rules in **Splunk**, reducing critical incident detection time by an average of **30 minutes**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)